CSCI3230 / ESTR3108 2026-27 First Term Assignment 1

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Question 1

The RSS objective is $J(\Theta) = \sum_{i=1}^{n} \big(f_\Theta(x^{(i)}) - y^{(i)}\big)^2$. Stacking the samples into a design matrix $\mathbf{X}$ (rows $1, x^{(i)}, \dots$) and the labels into $\mathbf{y}$, the minimizer solves the *normal equations* $\mathbf{X}^\top \mathbf{X}\Theta = \mathbf{X}^\top \mathbf{y}$.

(a) Analytic solution of $f_\Theta(x) = \theta_0 + \theta_1 x$

Here $\mathbf{X}$ has rows $\big(1,\, x^{(i)}\big)$, $i = 1, \dots, 10$. The needed sums are

$$\sum_i x^{(i)} = 56.1, \qquad \sum_i \big(x^{(i)}\big)^2 = 353.09, \qquad \sum_i y^{(i)} = 8.5, \qquad \sum_i x^{(i)} y^{(i)} = 52.88,$$

so the normal equations are

$$\begin{pmatrix} 10 & 56.1 \\ 56.1 & 353.09 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix} = \begin{pmatrix} 8.5 \\ 52.88 \end{pmatrix}.$$

The determinant is $\Delta = 10 \times 353.09 - 56.1^2 = 383.69 > 0$, hence the unique solution

$$\theta_1 = \frac{10 \times 52.88 - 56.1 \times 8.5}{383.69} = \frac{51.95}{383.69} = 0.1354, \quad \theta_0 = \bar{y} - \theta_1 \bar{x} = 0.85 - 0.1354 \times 5.61 = 0.0904.$$

Therefore

$$\boxed{f_\Theta(x) = 0.0904 + 0.1354\, x}, \qquad \text{training RSS} = 4.7816.$$

Figure 1 shows the fitted line together with the training points.

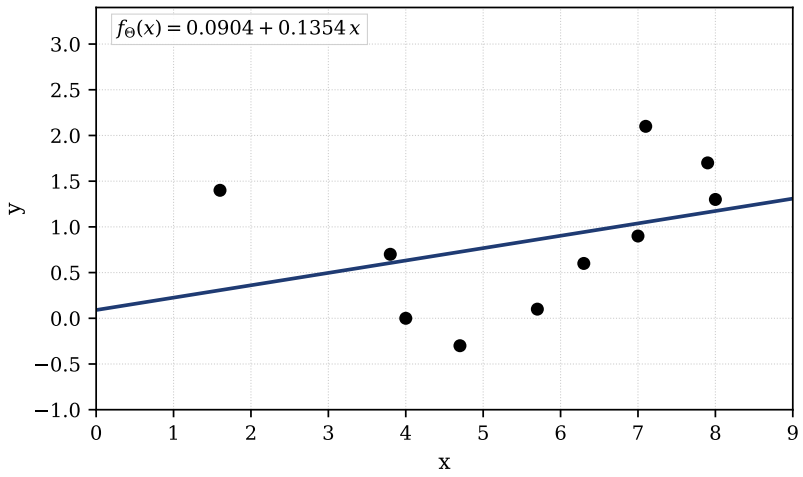


Figure 1: Training points and the fitted linear model $f_\Theta(x) = 0.0904 + 0.1354\,x$.

(b) Analytic solution of $f_\Theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$

Now the rows of $\mathbf{X}$ are $\big(1,\, x^{(i)},\, (x^{(i)})^2\big)$ and the normal equations form the 3×3 system

$$\begin{pmatrix} 10 & 56.1 & 353.09 \\ 56.1 & 353.09 & 2367.981 \\ 353.09 & 2367.981 & 16523.1077 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 8.5 \\ 52.88 \\ 373.386 \end{pmatrix},$$

where $\sum_i (x^{(i)})^3 = 2367.981$, $\sum_i (x^{(i)})^4 = 16523.1077$ and $\sum_i (x^{(i)})^2 y^{(i)} = 373.386$. Solving this system,

$$\boxed{f_\Theta(x) = 3.0589 - 1.2711\,x + 0.1394\,x^2}, \qquad \text{training RSS} = 1.8765.$$

Figure 2 shows the fitted parabola.

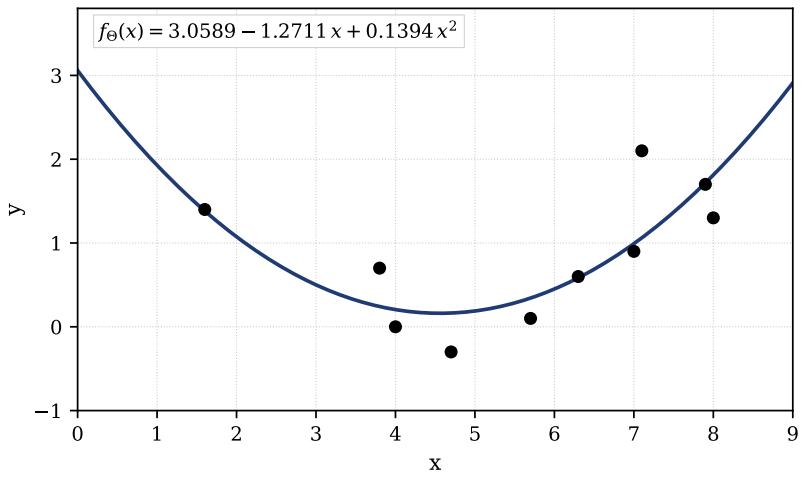


Figure 2: Training points and the fitted quadratic model $f_\Theta(x) = 3.0589 - 1.2711\,x + 0.1394\,x^2$.

(c) Analytic solution of the quartic model $f_\Theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

The rows of $\mathbf{X}$ are now $\big(1,\ x^{(i)},\ (x^{(i)})^2,\ (x^{(i)})^3,\ (x^{(i)})^4\big)$ and the normal equations form a 5×5 system

$$\begin{pmatrix} 10 & 56.1 & 353.09 & 2367.981 & 16523.1077 \\ 56.1 & 353.09 & 2367.981 & 16523.1077 & 118449.431 \\ 353.09 & 2367.981 & 16523.1077 & 118449.431 & 865703.6178 \\ 2367.981 & 16523.1077 & 118449.431 & 865703.6178 & 6418500.4415 \\ 16523.1077 & 118449.431 & 865703.6178 & 6418500.4415 & 48113659.523 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{pmatrix} = \begin{pmatrix} 8.5 \\ 52.88 \\ 373.386 \\ 2745.6248 \\ 20503.1486 \end{pmatrix},$$

where $\sum_i (x^{(i)})^k y^{(i)}$ for $k = 3, 4$ are 2745.6248 and 20503.1486. Solving,

$$\boxed{f_\Theta(x) = -12.1769 + 16.4458\,x - 6.3777\,x^2 + 0.9547\,x^3 - 0.0485\,x^4}, \quad \text{training RSS} = 0.7712.$$

Figure 3 shows the fitted curve together with the training points.

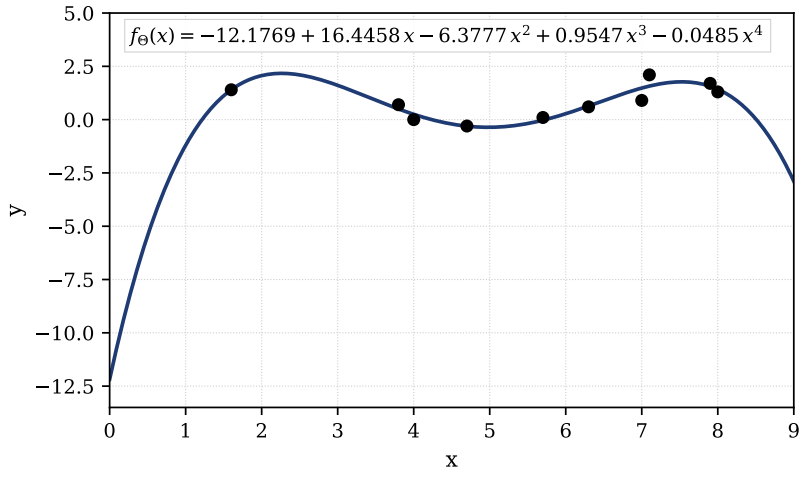


Figure 3: Training points and the fitted quartic model. The curve wiggles through the training points and takes extreme values outside the data range.

(d) Underfitting / overfitting and test-set verification

Qualitative comparison.

- **Degree 1 — underfitting.** The line cannot capture the curvature of the data (the dip around $x \approx 4.5$ is ignored). The residuals are systematically positive at the ends and negative in the middle; the model has high *bias* and its errors are large on *both* training and test sets.
- **Degree 4 — overfitting.** The quartic is flexible enough to nearly interpolate the 10 training points (smallest training RSS, 0.7712), but it bends aggressively between and beyond the points: e.g. $f_\Theta(0) = -12.18$ and $f_\Theta(9) = -2.89$, far outside the label range $[-0.3, 2.1]$. This high *variance* is not supported by the data, and indeed its test error is the *largest* of the three.
- **Degree 2 — relatively good.** The parabola captures the main curvature while keeping the model simple (three parameters). It attains the *smallest test RSS* (2.4768) with a still-moderate training RSS (1.8765): the best bias–variance trade-off of the three models.

Quantitative verification. Table 1 collects the training/test errors and Table 2 lists the per-point test predictions.

Model	Train RSS	Train MSE	Test RSS	Test MSE
Degree 1	4.7816	0.4782	4.1244	0.4124
Degree 2	1.8765	0.1877	**2.4768**	**0.2477**
Degree 4	**0.7712**	**0.0771**	5.1835	0.5184

Table 1: Training/test errors of the three models (4 d.p.). The smallest training error belongs to the quartic (overfitting), while the smallest test error belongs to the quadratic.

x	y	$f_1(x)$	$f_2(x)$	$f_4(x)$	e_1^2	e_2^2	e_4^2
7.2	0.6	1.0653	1.1333	1.6514	0.2165	0.2844	1.1055
1.1	1.0	0.2394	1.8294	−0.6040	0.5786	0.6879	2.5727
3.6	0.3	0.5779	0.2896	0.7699	0.0772	0.0001	0.2208
4.8	−0.4	0.7403	0.1694	−0.3381	1.3004	0.3242	0.0038
4.3	0.0	0.6726	0.1707	−0.0570	0.4524	0.0291	0.0033
6.6	0.2	0.9840	0.7418	1.0222	0.6147	0.2935	0.6760
7.8	1.8	1.1465	1.6251	1.6661	0.4270	0.0306	0.0179
5.6	0.6	0.8486	0.3123	−0.1109	0.0618	0.0828	0.5054
7.3	1.7	1.0788	1.2083	1.7122	0.3859	0.2418	0.0001
8.2	1.3	1.2007	2.0088	1.0209	0.0099	0.5024	0.0779
				Test RSS:	4.1244	2.4768	5.1835

Table 2: Test-set predictions and squared errors (f_1, f_2, f_4: degree 1/2/4; $e_k^2 = (f_k(x) - y)^2$; 4 d.p.). The quadratic achieves the smallest total test error; the quartic's largest errors occur at $x = 1.1$ and $x = 7.2$, where its oscillation hurts.

Comparing the residual magnitudes in Table 2: the degree-1 model is uniformly mediocre (large errors everywhere), the quartic achieves very small errors on most points but blows up on points $x = 1.1$, 7.2 (squared errors 2.57, 1.11), where its oscillation hurts, while the quadratic keeps all squared errors below 0.69. The numbers confirm the qualitative picture: **degree 1 underfits, degree 4 overfits, and degree 2 is the relatively good model**.

Question 2

Write $\mathbf{X}^{(i)}$ for the i-th sample (row vector), $\bar{\mathbf{X}} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{X}^{(i)}$ for the mean vector and consider

$$J(\Theta, \theta_0) = \sum_{i=1}^{m} \Big(\mathbf{X}^{(i)}\Theta + \theta_0 - Y^{(i)}\Big)^2 + \lambda\|\Theta\|^2, \qquad \lambda > 0.$$

(a) Proof that $\theta_0 = \bar{Y} - \bar{\mathbf{X}}\Theta$ minimizes J

Rewrite each squared term as

$$\mathbf{X}^{(i)}\Theta + \theta_0 - Y^{(i)} = \underbrace{\big(\mathbf{X}^{(i)} - \bar{\mathbf{X}}\big)\Theta}_{=:\,a^{(i)}} + \underbrace{\theta_0 + \bar{\mathbf{X}}\Theta}_{=:\,b} - Y^{(i)}.$$

For fixed Θ, differentiating w.r.t. θ_0 (note $\partial a^{(i)}/\partial\theta_0 = 0$ and $\partial b/\partial\theta_0 = 1$):

$$\frac{\partial J}{\partial \theta_0} = 2\sum_{i=1}^{m}\Big(a^{(i)} + b - Y^{(i)}\Big) = 2\Big(\sum_{i=1}^{m} a^{(i)} + mb - \sum_{i=1}^{m} Y^{(i)}\Big).$$

Since $\sum_{i=1}^{m}\big(\mathbf{X}^{(i)} - \bar{\mathbf{X}}\big) = \mathbf{0}$, the first sum vanishes:

$$\sum_{i=1}^{m} a^{(i)} = \Big(\sum_{i=1}^{m}\big(\mathbf{X}^{(i)} - \bar{\mathbf{X}}\big)\Big)\Theta = 0, \qquad \text{and} \qquad \sum_{i=1}^{m} Y^{(i)} = m\bar{Y},$$

hence

$$\frac{\partial J}{\partial \theta_0} = 2m\Big(\theta_0 + \bar{\mathbf{X}}\Theta - \bar{Y}\Big).$$

Setting $\partial J/\partial\theta_0 = 0$ gives

$$\theta_0 = \bar{Y} - \bar{\mathbf{X}}\Theta.$$

Moreover $\dfrac{\partial^2 J}{\partial \theta_0^2} = 2m > 0$ (the term $\lambda\|\Theta\|^2$ does not depend on θ_0), so J is a strictly convex quadratic in θ_0: this stationary point is the **unique global minimizer**. ■

(b) Substituting θ_0: $J(\Theta, \theta_0) = \tilde{J}(\Theta)$

Define the centered input $\tilde{\mathbf{X}}^{(i)} = \mathbf{X}^{(i)} - \bar{\mathbf{X}}$ and the centered label $\tilde{Y}^{(i)} = Y^{(i)} - \bar{Y}$. Substituting $\theta_0 = \bar{Y} - \bar{\mathbf{X}}\Theta$,

$$a^{(i)} + b - Y^{(i)} = \big(\mathbf{X}^{(i)} - \bar{\mathbf{X}}\big)\Theta + \bar{Y} - Y^{(i)} = \tilde{\mathbf{X}}^{(i)}\Theta - \tilde{Y}^{(i)}.$$

Therefore

$$J(\Theta, \theta_0) = \sum_{i=1}^{m}\Big(\tilde{\mathbf{X}}^{(i)}\Theta - \tilde{Y}^{(i)}\Big)^2 + \lambda\|\Theta\|^2 =: \tilde{J}(\Theta). \qquad \blacksquare$$

(c) Analytic solution of ridge regression

In matrix form, with $\tilde{\mathbf{X}}$ the matrix whose i-th row is $\tilde{\mathbf{X}}^{(i)}$ and $\tilde{\mathbf{Y}} = \big(\tilde{Y}^{(1)}, \ldots, \tilde{Y}^{(m)}\big)^\top$,

$$\tilde{J}(\Theta) = \big\|\tilde{\mathbf{X}}\Theta - \tilde{\mathbf{Y}}\big\|^2 + \lambda\|\Theta\|^2.$$

Using $\nabla_\Theta\big\|\tilde{\mathbf{X}}\Theta - \tilde{\mathbf{Y}}\big\|^2 = 2\tilde{\mathbf{X}}^\top\big(\tilde{\mathbf{X}}\Theta - \tilde{\mathbf{Y}}\big)$ and $\nabla_\Theta\|\Theta\|^2 = 2\Theta$,

$$\frac{\partial \tilde{J}}{\partial \Theta} = 2\tilde{\mathbf{X}}^\top\big(\tilde{\mathbf{X}}\Theta - \tilde{\mathbf{Y}}\big) + 2\lambda\Theta = 2\Big[\big(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda I\big)\Theta - \tilde{\mathbf{X}}^\top\tilde{\mathbf{Y}}\Big].$$

Setting the gradient to zero gives the *ridge normal equations*

$$\big(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda I\big)\Theta = \tilde{\mathbf{X}}^\top\tilde{\mathbf{Y}}.$$

Because $\lambda > 0$, for every $\mathbf{v} \neq \mathbf{0}$,

$$\mathbf{v}^\top\big(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda I\big)\mathbf{v} = \big\|\tilde{\mathbf{X}}\mathbf{v}\big\|^2 + \lambda\|\mathbf{v}\|^2 > 0,$$

so $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda I$ is positive definite and thus invertible. The analytic solution is

$$\boxed{\Theta^* = \big(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda I\big)^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{Y}}} \qquad \text{with} \qquad \theta_0^* = \bar{Y} - \bar{\mathbf{X}}\Theta^*. \qquad \blacksquare$$

Question 3

(a) Logistic model and cross-entropy error

With the linear model $f_\Theta(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ and $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2$, the logistic regression model is

$$P(\hat{y} = 1 \mid x_1, x_2) = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}, \qquad P(\hat{y} = 0 \mid x_1, x_2) = 1 - \sigma(z).$$

For the 6 training samples the cross-entropy (negative log-likelihood) error function is

$$E(\Theta) = -\sum_{i=1}^{6} \Big[y^{(i)} \ln \sigma\big(z^{(i)}\big) + \big(1 - y^{(i)}\big) \ln\big(1 - \sigma\big(z^{(i)}\big)\big) \Big], \qquad z^{(i)} = \theta_0 + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)}.$$

(b) One gradient-descent iteration

Since $\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$, differentiating E gives

$$\frac{\partial E}{\partial \theta_j} = \sum_{i=1}^{6} \Big(\sigma\big(z^{(i)}\big) - y^{(i)} \Big) x_j^{(i)}, \qquad x_0^{(i)} := 1.$$

With the initial parameters $(\theta_0, \theta_1, \theta_2) = (-1,\, 1.5,\, 0.5)$, Table 3 lists $z^{(i)}$ and $\sigma\big(z^{(i)}\big)$ for all training samples.

i	$x_1^{(i)}$	$x_2^{(i)}$	$y^{(i)}$	$z^{(i)}$	$\sigma\big(z^{(i)}\big)$
1	0.15	0.35	0	−0.6000	0.3543
2	0.30	0.45	0	−0.3250	0.4195
3	0.55	0.25	0	−0.0500	0.4875
4	0.30	0.80	1	−0.1500	0.4626
5	0.55	0.60	1	0.1250	0.5312
6	0.70	0.50	1	0.3000	0.5744

Table 3: Training samples with pre-activations and model outputs under the initial parameters (4 d.p.).

Writing $\sigma_i := \sigma(z^{(i)})$ and using the values of Table 3 (full precision carried in the computation):

$$\frac{\partial E}{\partial \theta_0} = \sum_i \big(\sigma_i - y^{(i)}\big) = \big(0.3543 + 0.4195 + 0.4875\big) - \big(0.5374 + 0.4688 + 0.4256\big) \approx -0.1705,$$

$$\begin{aligned} \frac{\partial E}{\partial \theta_1} = \sum_i \big(\sigma_i - y^{(i)}\big) x_1^{(i)} &= 0.3543(0.15) + 0.4195(0.30) + 0.4875(0.55) \\ &\quad - 0.5374(0.30) - 0.4688(0.55) - 0.4256(0.70) \approx -0.2698, \end{aligned}$$

$$\begin{aligned} \frac{\partial E}{\partial \theta_2} = \sum_i \big(\sigma_i - y^{(i)}\big) x_2^{(i)} &= 0.3543(0.35) + 0.4195(0.45) + 0.4875(0.25) \\ &\quad - 0.5374(0.80) - 0.4688(0.60) - 0.4256(0.50) \approx -0.4893. \end{aligned}$$

(The analytic gradient was verified against a central finite-difference estimate; the two agree to

within 4×10^{-10}.) The gradient-descent update $\Theta \leftarrow \Theta - \eta\,\nabla E$ with $\eta = 0.1$ gives

$$\begin{aligned}\theta_0 &= -1 - 0.1(-0.170474) = -0.98295,\\ \theta_1 &= 1.5 - 0.1(-0.269839) = 1.52698,\\ \theta_2 &= 0.5 - 0.1(-0.489345) = 0.54893.\end{aligned}$$

The updated logistic regression model is therefore

$$\boxed{P(\hat{y} = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-0.98295 + 1.52698\,x_1 + 0.54893\,x_2)}}}$$

and the cross-entropy error decreases from $E = 3.6077$ to $E = 3.5756$.

(c) Test-set predictions, accuracy, precision, recall

Applying the updated model to the test set (prediction rule $\hat{y} = 1$ iff $\sigma(z) \geq 0.5$, i.e. $z \geq 0$) gives Table 4.

i	$x_1^{(i)}$	$x_2^{(i)}$	$y^{(i)}$	$z^{(i)}$	$\sigma\big(z^{(i)}\big)$	$\hat{y}^{(i)}$
1	0.28	0.82	1	−0.1053	0.4737	0
2	0.45	0.75	1	0.1159	0.5289	1
3	0.58	0.62	1	0.2430	0.5605	1
4	0.72	0.48	1	0.3800	0.5939	1
5	0.22	0.40	0	−0.4274	0.3947	0
6	0.50	0.26	0	−0.0767	0.4808	0

Table 4: Test-set predictions of the updated model.

The confusion counts are: TP = 3 (samples 2–4), FP = 0, FN = 1 (sample 1), TN = 2 (samples 5–6). Hence

$$\text{Accuracy} = \frac{TP + TN}{6} = \frac{3+2}{6} = \frac{5}{6} \approx 0.8333, \qquad \text{Precision} = \frac{TP}{TP + FP} = \frac{3}{3+0} = 1.0000,$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{3}{3+1} = 0.7500.$$

Only sample 1 (true label 1, predicted 0 with probability $0.4737 < 0.5$) is misclassified. After a single gradient step the model already makes no false-positive errors (precision = 1), at the cost of one missed positive (recall = 0.75).

Question 4

Let $z_k = \mathbf{X}^\top \Theta_k$ for class k and write

$$P_i := P(\hat{y}_i = 1 \mid \mathbf{X}) = \frac{e^{z_i}}{\sum_{k=1}^{K} e^{z_k}}, \qquad i = 1, \ldots, K.$$

(a) Derivative of $P(\hat{y}_i = 1 \mid \mathbf{X})$ w.r.t. $\mathbf{X}^\top \Theta_j$

Case $j = i$: the numerator depends on z_i,

$$\frac{\partial P_i}{\partial z_i} = \frac{e^{z_i} \sum_k e^{z_k} - e^{z_i} e^{z_i}}{\left(\sum_k e^{z_k}\right)^2} = P_i - P_i^2 = P_i\big(1 - P_i\big).$$

Case $j \neq i$: the numerator does not contain e^{z_j},

$$\frac{\partial P_i}{\partial z_j} = \frac{0 \cdot \sum_k e^{z_k} - e^{z_i} e^{z_j}}{\left(\sum_k e^{z_k}\right)^2} = -P_i P_j.$$

In summary,

$$\frac{\partial P(\hat{y}_i = 1 \mid \mathbf{X})}{\partial\big(\mathbf{X}^\top \Theta_j\big)} = \begin{cases} P(\hat{y}_i = 1 \mid \mathbf{X})\big[1 - P(\hat{y}_i = 1 \mid \mathbf{X})\big], & j = i, \\ -P(\hat{y}_i = 1 \mid \mathbf{X})\, P(\hat{y}_j = 1 \mid \mathbf{X}), & j \neq i. \end{cases}$$

(Compactly: $P_i(\delta_{ij} - P_j)$, where δ_{ij} is the Kronecker delta.)

(b) Derivative of $P(\hat{y}_i = 1 \mid \mathbf{X})$ w.r.t. Θ_j

Since $z_j = \mathbf{X}^\top \Theta_j$, we have $\partial z_j / \partial \Theta_j = \mathbf{X}$, so by the chain rule

$$\frac{\partial P_i}{\partial \Theta_j} = \frac{\partial P_i}{\partial z_j} \cdot \frac{\partial z_j}{\partial \Theta_j} = \begin{cases} P_i\big(1 - P_i\big)\, \mathbf{X}, & j = i, \\ -P_i P_j\, \mathbf{X}, & j \neq i. \end{cases}$$

(c) Derivative of the negative log-likelihood loss

The loss is

$$E(\Theta) = -\sum_{i=1}^{K} y_i \ln P_i, \qquad P_i = P(\hat{y}_i = 1 \mid \mathbf{X}),$$

where exactly one y_i equals 1 (one-hot labels). Fix j and differentiate w.r.t. z_j:

$$\frac{\partial E}{\partial z_j} = -\sum_{i=1}^{K} \frac{y_i}{P_i} \cdot \frac{\partial P_i}{\partial z_j} = -\frac{y_j}{P_j} P_j\big(1 - P_j\big) - \sum_{i \neq j} \frac{y_i}{P_i}\big(-P_i P_j\big) = -y_j\big(1 - P_j\big) + P_j \sum_{i \neq j} y_i.$$

Because $\sum_{i=1}^{K} y_i = 1$, we have $\sum_{i \neq j} y_i = 1 - y_j$, so

$$\frac{\partial E}{\partial z_j} = -y_j + y_j P_j + P_j\big(1 - y_j\big) = P_j - y_j.$$

Finally, by the chain rule (as in part (b), with $\partial z_j / \partial \Theta_j = \mathbf{X}$):

$$\frac{\partial E(\Theta)}{\partial \Theta_j} = \frac{\partial E}{\partial z_j} \cdot \frac{\partial z_j}{\partial \Theta_j} = \big(P_j - y_j\big)\mathbf{X} = \mathbf{X}\, P(\hat{y}_j = 1 \mid \mathbf{X}) - \mathbf{X}\, y_j. \qquad \blacksquare$$